

Extending Reti-Pioneer with Monotone Quality Routing and Endpoint-Aware Multi-Task Learning

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Abstract

Screening endocrine and metabolic disease from retinal imaging remains limited by heterogeneous image quality, fragmented single-disease training loops, and risk scores that are seldom examined under calibration or decision-analytic criteria. Reti-Pioneer (Zhang et al., *Nature Medicine*, 2026) showed that frozen vision foundation models with quality-aware bilinear fusion can support multi-disease screening on UK Biobank (UKB) and hospital cohorts. The published article frames a multitask screening system. The released training entry point, however, optimizes independent binary heads in per-disease loops, and quality fusion uses fixed weights (good = 1, usable = 0.5, bad = 0).

We present a methods extension that preserves the frozen-backbone ensemble while revising three controllable components. First, quality scalars follow a monotone bounded route so that $\text{bad} \leq \text{usable} \leq \text{good}$ within $[0, 1]$, with good normalized to 1. Second, a shared multi-label head is trained with masked binary cross-entropy, supervising only labels present for each sample or dataset (endpoint-aware partial-label multi-task learning). Third, optional quality-conditioned backbone routing (configuration E5) maps the quality distribution to a softmax over foundation heads. Evaluation follows an endpoint-aware cross-cohort protocol on ODIR-5K, BRSET, and RFMiD, with alignment flags (direct / partial / related_not_equivalent), and reports expected calibration error (ECE), Brier score, and per-disease decision-curve analysis (DCA) after temperature scaling fitted on a fold disjoint from the scored set.

Clinical AUROC, ECE, and DCA tables on real foundation features remain 待补充. Section 5.1 reports pipeline-sanity metrics on SYNTHETIC feature caches computed in this repository (clinical_claim_allowed = false). These values must not be compared with Reti-Pioneer UKB internal AUROCs. Calibration and DCA constitute an evaluation framework rather than claimed algorithmic novelty.

Keywords: oculomics; fundus photography; monotone quality routing; partial-label multi-task learning; quality-conditioned backbone routing; calibration (evaluation); endpoint harmonization; Reti-Pioneer

1. Introduction

Color fundus photography is widely available in community eye care and can encode microvascular signatures of systemic disease. Foundation models such as RETFound provide transferable retinal representations, while quality assessment resources such as EyeQ / MCF-Net support handling of imperfect acquisitions. Reti-Pioneer integrated frozen RETFound, Swin Transformer V2-B, and Vision Mamba-S with quality-aware fusion and metadata, reporting internal-test AUROCs of 0.699-0.833 across six systemic conditions and describing a primary-care silent trial.

Three methodological gaps remain actionable without UKB access. First, Reti-Pioneer freezes

quality weights; unconstrained learnable linear weights can violate the clinical ordering of quality strata, so we constrain the learnable path to be monotone and bounded. Second, although the publication describes multitask screening, the released codebase trains independent binary heads; a shared multi-label head with masked supervision better matches multi-label public ocular datasets whose label spaces only partially overlap. Third, quality may also modulate which foundation heads should dominate the ensemble; we therefore provide an optional quality-conditioned backbone router (E5) as a gated extension. Deployment-oriented screening further needs calibrated probabilities and decision-curve net benefit. We treat temperature scaling, ECE, Brier, and DCA as a standard evaluation framework rather than as the paper's methodological novelty. Population-independent multi-disease studies further show that single-site AUROC overstates transportability when cameras, taxonomies, and populations shift.

Contributions.

1. A Reti-Pioneer-compatible monotone bounded quality router as the default learnable path, with fixed 1/0.5/0 retained as the released-code baseline and unconstrained free-linear routing as an ablation only.

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Boundary. We do not claim UKB type 2 diabetes mellitus (T2DM) AUROC 0.833 replication. We do not treat synthetic feature-cache AUROCs as clinical evidence. Gout, osteoporosis, hyperlipidemia, and thyroid remain UKB-only until public gold standards exist. ODIR diabetes (D) and BRSET diabetes labels are not UKB ICD endpoints. Over-broad diabetes_related mappings are exploratory only (related_not_equivalent).

2. Related work

Foundation models for color fundus photographs (CFPs). RETFound and related retinal foundation models reduce labeled-data needs for ocular and oculomic tasks. Reti-Pioneer ensembles multiple frozen backbones rather than proposing a new foundation model; our work inherits that stance.

Quality-aware analysis. EyeQ-style classifiers produce good / usable / reject (bad) probabilities. Reti-Pioneer's bilinear quality fusion is a strong baseline. Monotone routing is a bounded refinement of the fusion weights, not a new quality taxonomy. Quality-conditioned mixture over backbones is a complementary route that reallocates ensemble weight rather than only rescaling a fused scalar.

Multi-label fundus datasets and domain shift. ODIR-5K, RFMiD, and BRSET provide multi-label ocular (and some systemic) annotations downloadable by reviewers. They are not UKB ICD endpoints. We keep Reti-Pioneer's frozen-ensemble skeleton and change routing, head sharing, and endpoint-aware transfer reporting.

Calibration and decision curves (evaluation framework). Temperature scaling, ECE, and DCA are

standard clinical-utility assessments. We adopt them as reporting protocol, not as claimed algorithmic novelty.

Positioning. This is a methods paper on top of Reti-Pioneer, not a competitor foundation model and not a *Nature Medicine*-scale clinical claim without biobank data.

3. Methods

3.1 Problem formulation and inputs

Let each eye (or paired eyes) be represented by frozen backbone features $(x \in \mathbb{R}^d)$, a metadata vector (m) , and a three-way quality probability $(q = (q_{\text{good}}, q_{\text{usable}}, q_{\text{bad}}))$. Patient identifiers define splits so that both eyes of one person remain in one fold. When a calibration fraction is configured, we materialize train / calibration / validation / test so that temperature fitting never reuses the evaluation fold. RFMiD prefers official train / val / test CSVs when present.

3.2 Frozen backbones and feature caches

Following Reti-Pioneer, we freeze RETFound, Swin V2-B, and Vision Mamba-S and train fusion and heads on pre-extracted features (fast_mode). Public caches are built from prepared labels then foundation extraction. Until real pixels and CUDA extraction succeed, caches may contain deterministic synthetic features flagged by SYNTHETIC_FEATURES.txt with clinical_claim_allowed: false. Such caches support continuous integration only and are excluded from submission clinical tables.

3.3 Monotone bounded quality routing

Motivation. Fixed weights cannot adapt to site-specific quality distributions. Unconstrained linear weights can invert clinical orderings (for example, up-weighting “bad”). Monotonicity encodes the inductive bias that better quality should not receive a lower fusion weight than worse quality, while remaining differentiable end-to-end.

Mechanism. Quality-aware bilinear fusion maps features and a quality-derived scalar into a fused representation. We implement three routers:

Mode	Behavior
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For the monotone route we parameterize increments $(\delta \in \mathbb{R}^3)$, set $(u = \text{softmax}(\delta) + \epsilon)$, form $(c = \text{cumsum}(u))$, and obtain ordered weights $(w = c / c_3)$. Reordering aligns (w) with the (good, usable, bad) layout of (q) , yielding the scalar $(s = q_{\text{top}} w)$. The scalar enters the same bilinear fusion block used by Reti-Pioneer, so the architectural change is localized to the quality path.

Role in evaluation. Ablation arms contrast fixed versus monotone under otherwise matched training (configuration matrix E0-E4), isolating the effect of the ordering constraint from other factors.

3.4 Endpoint-aware partial-label multi-task learning

Motivation. Public ocular datasets expose partially overlapping label vocabularies. Independent

binary loops in the released training code cannot share representation across co-occurring labels when some endpoints are missing for a given sample or cohort.

Mechanism. `ComplexModel` exposes K logits. Masked BCE supervises only targets with a present label ($y \geq 0$); missing entries (-1 / NaN) are ignored in the loss denominator. This supports joint training across datasets whose label vocabularies only partially overlap (for example, ODIR eight-way ocular codes versus BRSET systemic / ocular fields), including an optional multi-cohort joint vocabulary with per-endpoint masks.

The released-code control retains independent single-disease training loops as in upstream `main.py`. Ensemble modes include `released_code` (soft vote in training / max at inference), `published_soft_vote`, `mean`, and `temperature-averaged` variants. Legacy `ensemble="paper"` aliases to `released_code` with a deprecation warning to avoid conflating publication wording with training mechanics.

Role in evaluation. Multitask versus single-task arms (E2-E4) hold the quality router fixed or jointly varied, so masked supervision can be attributed separately from monotone routing.

3.5 Optional quality-conditioned backbone routing (E5)

Motivation. Uniform mean or fixed soft-vote ensembles ignore that different foundation heads may be more reliable under poor acquisition. Attenuating features by a gate can hide that effect inside the fusion block; an explicit mixture over heads makes the reallocation inspectable.

Mechanism. When enabled (`quality_gating`), a small multilayer perceptron maps q to a softmax over the three foundation heads, replacing uniform mean or fixed soft-vote with a quality-conditioned mixture (`QualityBackboneRouter`). An optional soft-quality auxiliary loss (λ_q) can be used when soft quality targets are available (for example, BRSET). E5 is config-gated (`configs/ablation_e5.yaml`: `quality_gating: true`, `lambda_q: 0.1`, `monotone + multitask + masked BCE`) and is not asserted as a clinical improvement in this draft. A legacy feature-attenuation gate remains available for ablation only.

Role in evaluation. E5 is reported as an optional arm once real foundation features are available; synthetic pipeline tables in Section 5.1 focus on E0-E4-style baseline / `learnq` / `multitask` / `full` runs that completed under CPU feature caches.

3.6 Calibration and decision utility (evaluation framework)

Temperature T is fit on `calibration_ids` disjoint from `evaluation_ids`. If a request would fit and score the same set, `paper_mode` errors; otherwise a held-out calibration slice is nested and a warning is emitted. Operating points (Youden / sensitivity at 95% specificity) are selected on validation or calibration only and applied frozen on the evaluation fold. We report AUROC / average precision with optional patient-level bootstrap 95% CI and DeLong asymptotic CI, equal-width ECE, Brier score, optional calibration intercept / slope, sensitivity at high specificity, per-disease DCA curves, and (illustrative only) net benefit at threshold 0.10 versus `treat-all` / `treat-none`.

3.7 Training protocol

| Item | Setting |

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3.8 Endpoint ontology (pre-registered)

Endpoint · Kind · ODIR · BRSET · Alignment notes

Cross-evaluation with clinical-table or paper-mode guards requires alignment=direct. Full harmonization lives with the public-data documentation and label map.

3.9 Failure modes and out-of-scope

Missing quality vectors are padded with a loader-defined default. Vision Mamba is optional when wheels or weights are absent. Synthetic caches must never populate submission clinical tables. UKB-only diseases are omitted from public tables. BRSET requires PhysioNet credentialing. Extension inference uses the project predict entry point rather than upstream image-API inference when only caches are available.

4. Experiments (protocol)

- Datasets: ODIR-5K, BRSET, RFMiD (full-image download and three-backbone extraction status: see Data availability; clinical tables 待补充).

- Baseline
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Synthetic or demo AUROC must never appear in submission clinical tables without an explicit SYNTHETIC label and clinical_claim_allowed = false.

5. Results

5.1 Pipeline verification on SYNTHETIC feature caches (not clinical)

The following metrics were computed in this repository by scripts/run_ablations.py (timestamp 2026-09-15T02:45:26) on deterministic synthetic ODIR / BRSET feature caches (SYNTHETIC_FEATURES.txt; evaluate JSON disclaimer: clinical_claim_allowed=false unless alignment=direct). Sample sizes are small (ODIR val \ (n=10\); ODIR→BRSET \ (n=48\)). Values near chance are expected. Arm ranking must not be interpreted as method superiority, and numbers must not be compared with Reti-Pioneer UKB AUROCs (0.699-0.833).

Table 1. Ablation summary (SYNTHETIC; repository-computed metrics). Primary comparable column:

AUROC\(_D\)

 (ODIR D head). Macro AUROC is multi-label for multitask arms.

Arm · Eval · AUROC · AUROC\(_D\)

 · ECE · Cal. ECE · \(\mathcal{T}\) · \(\mathcal{N}\)

Temperature scaling changes ECE without changing ranking AUROC, which is the intended behavior of a post-hoc calibrator. Cross-cohort columns illustrate the protocol of reporting domain drop under endpoint mapping, not a clinical transportability estimate. Extremely small \(\mathcal{N}\) produces unstable point estimates (for example, AUROC = 1.0 on some ODIR heads in the full-arm JSON); such values are artifacts of synthetic labels and tiny folds.

Companion figures (SciencePlots, Times New Roman): architecture schematic (Figure 1); SYNTHETIC ablation bars (Figure 2); ECE before / after temperature scaling (Figure 3); cross-cohort AUROC\(_D\) drop (Figure 4). Provenance is recorded in docs/paper/AUTHENTICITY_AUDIT.md.

5.2 Clinical evaluation on real foundation features

待补充. Requires (a) authorized download of ODIR / BRSET / RFMiD images under their licenses, (b) CUDA-capable extraction of RETFound / Swin / Vim features, and (c) multi-seed runs with paper_mode guards. RFMiD images and official CSVs are present in this workspace (\(\mathcal{N}=3200\)), but three-backbone foundation features for clinical tables remain unavailable on CPU-only PyTorch. ODIR (Kaggle) and BRSET (PhysioNet) downloads were blocked without credentials at drafting time.

6. Discussion

Public ocular and systemic labels enable reproducible methods claims that UKB-gated papers cannot always support. Transferring Reti-Pioneer's skeleton to ODIR / BRSET / RFMiD forces explicit endpoint harmonization and domain-shift reporting. Fairness and calibration should be first-class evaluation practices, matching the authors' own limitations discussion, without overselling calibration or DCA as novelty.

Figure 1 is an information-flow schematic (yellow = monotone quality fusion; green = partial-label multi-task head and calibration / DCA evaluation). Figures 2-4 currently plot SYNTHETIC feature-cache metrics and justify engineering readiness of the protocol, not clinical superiority over Reti-Pioneer's UKB AUROCs.

A concise novelty statement is: we replace fixed quality weights with a monotone bounded router, train a masked multi-task head against the released independent-loop control, optionally route backbone mixture weights from quality (E5), and evaluate with endpoint-aware cross-cohort splits plus a calibration / DCA reporting framework. We do not claim to outperform Reti-Pioneer, and we do not report a UKB T2DM AUROC without the original endpoint and cohort.

Open risks for subsequent clinical tables include empty real-feature Results, published-versus-released multitask wording, endpoint drift if diabetes_related is misused clinically, monotone weights that may remain near (0, 0.5, 1) without quality supervision,

illustrative DCA at threshold 0.10, and blocked foundation extraction without CUDA.

7. Limitations

No UKB / SEED access in this drafting workspace; label mismatch versus ICD systemic endpoints; no prospective trial; Vision Mamba optional on some platforms; EyeQ weights external; synthetic caches for continuous integration only; ODIR Kaggle credentials absent; BRSET PhysioNet data-use agreement not completed here; GPU absent for RETFound / Swin extraction. Clinical AUROC tables therefore remain 待补充.

8. Reproducibility and availability

Code, ablation configs, patient-level splits with calibration hashes, and evaluation guards are released for independent checking. A public code-and-docs snapshot is available at <https://github.com/Coucou2016/retinal-imaging-methods-public> (no patient images or UKB extracts). See third-party notices for upstream Reti-Pioneer attribution.

Data availability. Public ODIR-5K / BRSET / RFMiD under their respective licenses (BRSET via PhysioNet credentialing). UK Biobank data are not redistributed; clinical UKB AUROCs remain 待补充 until authorized access and re-analysis.

Representative commands (environment-dependent):

text

pytho

References

1. Zhang, X. et al. AI framework for multidisease detection via retinal imaging. *Nat Med* 32, 2494–2503 (2026). doi:10.1038/s41591-026-04359-w

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AUROC

